Cleaned_Omair_Sales_Data.csv

InsightFlow AI analysis report · Sales dataset · generated August 05, 2026

Executive Summary

The sales dataset indicates a healthy business model with over $10 million in revenue and a strong margin profile, though profitability is highly sensitive to fluctuations in supply costs. While the overall data quality is excellent, significant multi-year gaps in reporting prevent reliable long-term trend analysis and limit our ability to benchmark current performance against historical data. To improve strategic planning, we must investigate recurring seasonal slumps in June and November and isolate cost outliers to ensure our margin analysis remains accurate.

Key Metrics

Metric	Value
Total Revenue	10075360.98$
Total Orders	1998
Average Transaction Value	5042.72$
Data Quality Score	99.96%

Insights

Strong Cost-to-Sale Correlation [INFO]

Sales and costs maintain a strong correlation of 0.86, indicating that revenue growth is heavily dependent on underlying cost inputs. Since the mean sale ($5,042) is significantly higher than the mean cost ($3,149), the business is maintaining a healthy margin; however, any volatility in supply costs will likely impact bottom-line profitability proportionately.

Recurring Seasonal Performance Dips [WARNING]

Performance data shows recurring transaction volume drops in November (59 transactions in 2004, 72 in 2005) and June (42 in 2004, 71 in 2005, 43 in 2006) relative to neighboring months. Investigating these localized slumps could reveal if these are caused by inventory stock-outs, external competitive pressure, or specific regional shopping habits during these times.

Significant Data Fragmentation [CRITICAL]

The dataset contains massive temporal gaps, including a 98-month absence between June 2006 and August 2014, and a 16-month gap leading into December 2015. These discontinuities render longitudinal trend analysis unreliable, as current performance cannot be accurately compared to historical baselines without accounting for these long periods of inactivity.

Outlier Impact on Cost Reporting [WARNING]

While the dataset maintains an excellent quality score of 99.96%, the presence of 5 outliers in the 'Cost' column suggests individual transactions that deviate from the norm. Given that the standard deviation for cost is quite high ($2,057), these outliers may be skewing mean cost figures and should be isolated to ensure accurate margin analysis.

Data Cleaning

- **Cost:** 5 IQR-based outlier value(s) found and left in place (not removed automatically).

Forecast

Not enough complete time periods (need at least 4, found 1) after excluding 26 earlier period(s) separated from the recent data by a large gap